

Belief-State Safety Values for Viability under Incomplete Observation

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Abstract

The obstruction calculus characterizes nonviability through certificates on information states; the probabilistic side asks what the certificates imply when the information state is a distribution. This paper develops the belief-state safety value $V_k(b)$ — the maximal probability of remaining safe for k steps under the disturbance’s worst case — and establishes four results. First, a support identity: on finite models with deterministic kernels the posterior support coincides with the set-valued post-state of the calculus, the value-one level of the recursion coincides with the viable-set recursion $\mathcal{W}_k$, and outside it the deficit satisfies $1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x)$, attained on natural instances. Second, exactness: over rational input data the finite-horizon values are piecewise linear with rational alpha-vectors, so all values are exact rationals computable without floating point; the verification campaign’s 384 vectors are all exact. Third, agreement on the audited delayed-regime class: a class-restricted alpha-vector recursion reproduces the closed-form values on all 48 audited cells and horizons $k \leq 4$, the certificate partition coincides with the value boundaries, and the value jumps by exactly $1/2$ at $z_0 = 1 + \min(k, T_{\text{obs}})$; the declared hold class equals the unrestricted sequential blind class at the one-step horizon, and for horizons $k \geq 2$ the two differ exactly on the timing cells ($2 \leq z_0 < 1 + T_{\text{obs}}$), where open-loop time-variation within the blind window lifts the value to one. Fourth, adaptive observation: on the declining hidden-parameter instance a single probe reveals the parameter exactly (branch separation 2), survival forces $z_0 \geq 21/10$, and under revelation at horizon T_{learn} viability holds exactly when $z_0 \geq 21/10 + T_{\text{learn}}/10$ — the learning deadline acts as the timing obstruction’s observation deadline, and the kernel shrinks linearly and antitone in it ($|K| = 5, 4, 3, 2$). All computations run in exact rational arithmetic.

Keywords: partially observable control; viability kernels; belief states; safety values; exact arithmetic; adaptive observation

1. Introduction

Viability under incomplete observation divides into a necessity side and a sufficiency side. The obstruction calculus treats the necessity side: certificates whose firing proves nonviability of an information state (Abaee,

2026b). The probabilistic question is the quantitative counterpart. If the controller holds a distribution b over states rather than a set, how safe is safe? The answer is a function: the belief-state safety value $V_k(b)$, the maximal worst-case probability of keeping the state trajectory in the safe set for k steps. Its value-one level is the viability statement of the set-valued theory; everything below level one is measured by the deficit $1 - V_k(b)$, and the obstruction certificates reappear as lower bounds on that deficit.

This paper develops the value function in four steps. Section connects the belief-state recursion to the set-valued recursion of the calculus exactly, through the support of the belief. Section shows the finite-horizon values are piecewise linear and exactly representable. Section evaluates the deterministic limit, at which the recursion degenerates to a survived-mass formula over blind action sequences. Section reads the obstruction certificates systematically as deficit lower bounds. Section proves agreement between the class-restricted value recursion and the closed-form values on the audited delayed-regime class, quantifies what the class declaration contributes, and locates the value’s jump loci. Section treats observation as a design variable: probes, learning deadlines, and the exact kernel cost of late resolution of a hidden parameter.

All audited computations are exact: integer and rational arithmetic, standard library only, deterministic scripts (Section).

The belief-state model and the value recursion

The model is the calculus’s finite model with an absorbing unsafe state. Let X be a finite state set with a designated safe set $\mathcal{V} \subseteq X$, A a finite action set, Y a finite observation set, and D a finite disturbance set with deterministic transition and observation maps $x^+ = F(x, a, d)$ and $y = O(x, a, x^+)$; total transition rows over the augmented space $X \cup \{\partial\}$, with ∂ absorbing, unsafe, and carrying its own observation, convert set-valued nonviability into almost-sure nonviability on the augmented chain. A belief is a probability distribution b on X ; the posterior after action a and observation y is $b^+(x') \propto \sum_x b(x) \mathbb{1}[y = O(x, a, x')]$.

Definition 1 (safety value). For horizon k and belief b ,

$$V_k(b) = \sup_{\pi} \inf_{d\text{-strategies}} \mathbb{P}_b^{\pi}(x_t \in \mathcal{V} \text{ for all } t \leq k),$$

the supremum over observation-based policies and the infimum over disturbance strategies.

Theorem 2 (value recursion). $V_0(b) = \mathbb{K}[b \in \mathcal{P}(\mathcal{V})]$, and for $k \geq 1$,

$$V_k(b) = \sup_a \mathbb{E}_{y \sim b} V_{k-1}(b^+(\cdot | a, y)),$$

where the expectation is over the observation distribution induced by b and a . The operator is the robust dynamic program on beliefs; V_k is concave and piecewise linear in b whenever the one-step data are.

The support identity and exact sufficiency

The bridge to the set-valued theory runs through the support. Write $\text{supp}(b)$ for the states carrying positive mass, and let $\mathcal{W}_k$ denote the calculus's viable-set recursion: $\mathcal{W}_0 = \{B \subseteq \mathcal{V}\}$ and $\mathcal{W}_k = \{B : \text{some admissible } a \text{ maps every } x \in B \text{ into } \mathcal{W}_{k-1}\text{-supported posteriors}\}$.

Theorem 3 (support identity). *On finite models with deterministic kernels, for every action a and observation y ,*

$$\text{supp}(b^+(\cdot | a, y)) = \text{Post}(\text{supp}(b), a, y),$$

the set-valued post-state of the calculus. Consequently $V_k(b) = 1$ if and only if $\text{supp}(b) \in \mathcal{W}_k$, while if $\text{supp}(b) \notin \mathcal{W}_k$ then every action loses a compatible state of positive mass, so

$$1 - V_k(b) \geq \min_{x \in \text{supp}(b)} b(x).$$

Moreover $V_k(b)$ is non-increasing in k .

Corollary 4 (closed form on the delayed class). *On the delayed hidden-regime instance — states (z, θ) with $\theta \in \{-1, +1\}$, prior $b_0 = (\frac{1}{2}, \frac{1}{2})$ from z_0 , no observation before T_{obs} , exact revelation at T_{obs} , blind-window holds $u \equiv \pm 1$, and unit drift against the floors — the belief support follows the two branches $z_0 \mp k$, and*

$$V_k(b_0) = \begin{cases} \frac{1}{2} (\mathbb{K}[z_0 - k \geq 1] + \mathbb{K}[z_0 + k \geq 1]), & k \leq T_{\text{obs}}, \\ \frac{1}{2} (1 + \mathbb{K}[z_0 \geq 1 + T_{\text{obs}}]), & k > T_{\text{obs}}. \end{cases}$$

In particular $V_k(b_0) = 1$ exactly on the viable region $z_0 \geq 1 + T_{\text{obs}}$, recovering the deterministic verdict as the value-one level.

Exact piecewise-linear values

Proposition 5 (rational alpha-vectors). *Over rational input data, the backward recursion of Theorem 2 propagates finite sets of rational alpha-vectors: each V_k is the maximum of finitely many rational affine functions of b , every value is an exact rational, and no floating point is required at any horizon. In the verification campaign the alpha-vectors of all audited instances — 384 vectors across the two-floor instance and the delayed grid — are exact rationals.*

Remark 6 (dimension). On the two-floor instance the value function on the one-dimensional simplex is $\max(b_0, b_1)$ for all audited horizons; the witness alpha-vectors live in the two-coordinate display, with the suppressed bottom coordinate identically zero.

The deterministic limit

As the observation vanishes the value recursion is replaced by a distinct robust operator.

Proposition 7 (survived-mass formula). *On the delayed class with blind window T_{win} , the k -step value of the prior equals the maximal survived mass over blind action sequences:*

$$V_k(b_0) = \max_{u_{1:k}} \left[w \mathbb{K}[z_0 + \sum_{t \leq k} u_t \geq 1] + (1 - w) \mathbb{K}[z_0 - \sum_{t \leq k} u_t \geq 1] \right],$$

with $w = \frac{1}{2}$ on the symmetric prior and the supremum attained on pure sequences. This is a different operator from the robust verdict recursion of the deterministic theory, and it recovers that theory's verdicts as its 0/1 specialization.

On the two-floor chance instance — states $\{0, 1\}$, actions $u \in \{\frac{2}{5}, \frac{3}{5}\}$, state 0 surviving exactly the lower action and state 1 exactly the higher — the one-step servable sets are $\Gamma_1 = \{(1, 0), (0, 1)\}$ and $V_k(b_0) = \frac{1}{2}$ for all audited $k \leq 5$: the chance deficit $\delta = \frac{1}{2}$ is attained, and the deficit equals the minimal mass.

Certificates as deficit bounds

The obstruction certificates bound the deficit from below.

Proposition 8 (deficit lower bounds). *The following hold on the audited instances. (i) Chance-constrained common-action bound: if no single action keeps all compatible states safe, the deficit is at least the minimal compatible mass. (ii) Degenerate minimal-mass bound: in the deterministic limit the deficit is at least the minimal mass of the belief, attained whenever exactly one branch is savable (the two-floor instance attains $\delta = \frac{1}{2}$; on the delayed grid at $(z_0, T_{\text{obs}}) = (\frac{3}{2}, 2)$ every blind sequence loses a branch, so the deficit is at least the minimal mass $\frac{3}{10}$ of the audited asymmetric prior). (iii) Monotone deficit: the deficit is non-decreasing in the horizon on the audited class, and on the certified nonviability region of the delayed grid the deficit is constant.*

Agreement on the delayed-regime class

The class declaration is part of the model: the blind-window policy may be required to hold one action throughout the window (the declared hold class), or may act sequentially without observation, or may act with open-loop time-variation.

Theorem 9 (certificate-value agreement). *On all 48 audited cells (z_0, T_{obs}) with $z_0 \in \{1.0, 1.1, \dots, 2.5\}$ and $T_{\text{obs}} \in \{1, 2, 3\}$, and horizons $k \leq 4$: (i) the class-restricted alpha-vector recursion of Proposition 5 reproduces the closed form of Corollary 4; (ii) the certificate partition of the grid coincides with the value boundaries; (iii) the attaining witness and the value's jump loci trace the certificate boundary exactly, the jump being exactly $\frac{1}{2}$ at $z_0 = 1 + \min(k, T_{\text{obs}})$ and the value constant on either side.*

Theorem 10 (class declaration). *The declared hold class and the unrestricted sequential blind class coincide at the*

one-step horizon. For horizons $k \geq 2$ they differ exactly on the timing cells — those with $2 \leq z_0 < 1 + T_{\text{obs}}$ — and there open-loop time-variation within the blind window lifts the value to one: an alternating hold sequence keeps both branches at or above the floor whenever $z_0 \geq 2$, which no constant hold does. On the common-action cells $z_0 < 2$ the restriction costs nothing — no blind sequence saves both branches — and on the viable cells both classes attain one.

Adaptive observation and hidden parameters

Observation is a design variable. On the declining instance the state carries a hidden parameter $\theta \in \{-1, +1\}$ gating the correctable component of a $-\frac{1}{10}$ drift, and the belief lives on the product (z, θ) ; the obstruction is the emptiness of the joint floor-servable set over (x, θ) cells.

Proposition 11 (probes and learning deadlines). *On the audited grid: with θ observed, every cell is viable (16 of 16). Under one-probe endogenous learning, the probe reveals θ exactly — the branch separation is 2, so one observation splits the belief with no statistics — but survival forces $z_0 \geq 21/10$. Under exogenous revelation at horizon T_{learn} , viability holds exactly when*

$$z_0 \geq \frac{21}{10} + \frac{T_{\text{learn}}}{10}, \quad T_{\text{learn}} = 0, 1, 2, 3, 4,$$

the learning-deadline identity: late resolution acts as the timing obstruction’s observation deadline, and the kernel shrinks exactly linearly and antitone in it, $|K| = 5, 4, 3, 2$ on the grid for $T_{\text{learn}} = 0, 1, 2, 3$. A two-patch instance with unknown regeneration rate $g \in \{1, 2\}$ shows the same phenomenon on the audited two-patch belief pair: unknown-regeneration nonviable, (x, g) -observed viable.

Conclusion

The belief-state safety value carries the obstruction calculus into the probabilistic setting without loss of exactness. The support identity ties the value-one level to the viable-set recursion; rational alpha-vectors keep every audited value exact; the certificates survive as deficit lower bounds, attained on natural instances; the audited delayed-regime class agrees with its closed form cell by cell, with the class declaration responsible for exactly the timing cells; and the adaptive-observation audit prices late resolution exactly — a linear learning deadline on the viable set. The observation layer, not the forecast layer, sets the value; the design rules are the same ones the deterministic audits impose, now with a price attached.

Verification methods

Every value, bound, identity, and threshold in this paper is regenerated by deterministic verification scripts in exact integer and rational arithmetic, standard library only. Two per-system scripts cover the audited instances — fifteen check families on the stochastic values (the two-floor instance, piecewise-linearity, the deterministic limit, the 48-cell agreement, the partition coincidence, the witness family, the deficit identity, the unrestricted class, rationality of all 384 campaign alpha-vectors, deficit monotonicity,

the oscillation identity, model completeness, the selector split, dimension consistency, and the jump loci) and six check families on the adaptive-observation instance (theta observed, the probe threshold, the learning-deadline identity, injective revelation, the antitone kernel chain, and the unknown-regeneration instance). A master script chains the two, then recomputes independently from the definitions: the closed-form grid values are reproduced by direct branch-mass enumeration for every cell with $k \leq T_{\text{obs}}$, the minimal-mass bound is verified attained on the two-floor instance, the jump loci are recomputed from the closed form on the grid, and every headline number is asserted present in the text.

References

Abaee, A. An obstruction calculus for viability under incomplete observation (submitted).

Abaee, A., 2026a. Robust viability of the 2J3KL limit reference point under a surplus-production map: policy scoring, expansion, and when catch cannot help. Zenodo. <https://doi.org/10.5281/zenodo.22552060>

Declarations

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Competing interests

None.

Data availability

No external data were used.

Code availability

The verification scripts `paper2_belief_state_v1_verification.py`, `paper2_stochastic_selector_v2_verify.py`, and `hidden_parameter_learning_v1_verify.py` are publicly available at <https://github.com/MIKEAA2020/general-sustainability> (folder `arena agent 1/paper rewrites/latex`); they reproduce all values, bounds, and thresholds verbatim.

AI declaration

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